

Wine Quality Prediction



- ▶ Final Project for AI240: Machine Learning, Bellevue College – Spring 2025
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Problem Statement

- ▶ The goal of this project is to build a machine learning model that predicts the **quality score of red wine (from 3 to 9)** based on its **chemical properties**, such as acidity, sugar content, pH, and alcohol level.

What Is Achieved

By completing this project, we can:

- ▶ Understand which chemical features influence wine quality.
- ▶ Build a tool to predict wine quality based on measurable properties.
- ▶ Compare the performance of different machine learning models

Why It Is Important

- ▶ Wine quality is typically judged by human experts, which can be time-consuming and subjective. With machine learning, we can automate this process, **making it faster, more consistent, and data-driven.**

Is it a Classification or Regression ML Project?

Why This Is a Classification, Not Regression

- ▶ Wine quality is a **label**, not a continuous number.
- ▶ **Classification**: predicts categories like 3, 4, 5... not “how close.”
- ▶ **Regression**: predicts values like 6.43 and cares about how close the prediction is.

Our project: wine quality is one of 7 discrete classes (3–9), so it’s a **multi-class classification** problem.

Target and Categories

- ▶ Target: Wine quality (integer from 3 to 9)
- ▶ Features: 11 chemical attributes:
 - ▶ Fixed Acidity
 - ▶ Volatile acidity
 - ▶ Citric Acid
 - ▶ Residual Sugar
 - ▶ Chlorides
 - ▶ Free Sulfur Dioxide
 - ▶ Total Sulfur Dioxide
 - ▶ Density
 - ▶ pH
 - ▶ Sulphates
 - ▶ Alcohol

Dataset Used

- ▶ Balanced version of UCI Wine Quality dataset from Kaggle.
- ▶ Link: <https://www.kaggle.com/datasets/taweilo/wine-quality-dataset-balanced-classification>
- ▶ Why I Chose the Balanced Wine Dataset

The original UCI Wine Quality dataset has a highly imbalanced distribution — most wines are rated 5 or 6, and very few are rated 3 or 8. This makes it harder for a machine learning model to learn meaningful patterns for underrepresented classes.

- ▶ In the **balanced version of the dataset**, all wine quality scores (3–9) appear in equal amounts. This allowed me to:
 - Train a fair model without adding bias
 - Use standard evaluation metrics like accuracy more reliably
 - Focus on comparing different classification algorithms without needing extra steps like oversampling

ML Algorithms Used

► **Logistic Regression**

- A good starting model for classification problems
- Simple and fast to train
- Helps see how well simple patterns work on this data

► **Decision Tree**

- Handles more complex patterns and rules
- Can work better when features combine in tricky ways
- Easy to explain and show how the model makes decisions

ML Libraries Used

- ▶ **pandas** – for loading and working with the dataset
- ▶ **matplotlib.pyplot** – for basic plotting
- ▶ **seaborn** – for nicer-looking charts and heatmaps
- ▶ **sklearn.model_selection** – to split data into training and testing sets
- ▶ **sklearn.linear_model** – to use Logistic Regression
- ▶ **sklearn.tree** – to use Decision Tree Classifier
- ▶ **sklearn.metrics** – to check model accuracy and show confusion matrix

Evaluation Criteria

► Accuracy Score

Tells us the overall percentage of correct predictions.

Simple, but may not tell the full story in multi-class problems.

► Confusion Matrix

Shows where the model made mistakes.

Helps us see which quality scores are most confused.

► Classification Report

Shows how well the model performs for each class using:

Precision, **Recall**, and **F1-score** — these help measure accuracy, coverage, and overall balance.

Model Performance

- ▶ Logistic Regression: 29% accuracy
- ▶ Decision Tree: 59% accuracy

Model Comparison

► **Logistic Regression** – Accuracy: **29%**

- Works best for simple, straight-line patterns
- Wine quality depends on many features working together — too complex for this model
- It predicted class 5 and 6 fairly well, but struggled with the rest

► **Decision Tree** – Accuracy: **59%**

- Handles non-linear patterns and feature combinations
- Can learn rules like: “If alcohol is high and acidity is low → quality is 7”
- Performed more evenly across all classes

Conclusion

- ▶ Wine quality depends on multiple interacting features.
- ▶ Decision Tree was the better choice for this task.
- ▶ The project shows how ML can help in real-world quality control. Even though 59% isn't perfect, it's reasonable for a 7-class prediction problem.